

Leonardo Bonacini

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Career Objective

Looking for a **Robotics Engineer, Autonomy Engineer or Perception/Navigation Engineer** role. For the past four years I have worked on navigation software for field robots used commercially in Brazil and the US: perception, sensor fusion, localization, mapping, planning and control, in ROS/ROS 2, C++ and Python on NVIDIA Jetson. I have recently started working with LLMs, applied to fault diagnosis.

Professional Experience

- **University of São Paulo** **São Carlos - Brazil**
Postdoctoral Researcher *2026–present*
Research on autonomous robot navigation for companies in the oil and gas sector.
 - Development of perception, localization and trajectory planning algorithms for robots in industrial sites.
 - Started a project on using LLM agents for bearing fault detection.
- **EarthSense, Inc.** **Champaign-Urbana - USA**
Senior Robotics Engineer *2023–2026*
Developed the autonomy stack for agricultural robots used in commercial operations in Brazil and the United States.
 - Navigation software from perception to control: sensor fusion (LiDAR, IMU, GNSS/RTK and stereo vision), localization, mapping, trajectory planning and real-time control, with ROS/ROS 2, C++ and Python.
 - YOLO-based perception for obstacle detection and traversable-area segmentation, running on NVIDIA Jetson with TensorRT.
 - Collaboration with international development teams (USA, India); used LLMs to organize project documentation and onboarding material for new engineers.
- **São Carlos School of Engineering - University of São Paulo** **São Carlos - Brazil**
Agricultural Robotics Researcher *2022–2023*
Developed the autonomy software for an agricultural robot designed for targeted pesticide application in cotton and row crops in the state of Mato Grosso, Brazil. Developed perception, localization, mapping, trajectory planning, obstacle avoidance and motion control algorithms using ROS, C++, Python, LiDAR, GNSS/RTK, IMU, stereo vision and deep learning techniques, enabling reliable autonomous operation in real agricultural environments.

- **Claretiano - Centro Universitário de Batatais** **Batatais - Brazil**
Educational Content Developer 2022
 Developed course material for the Mechanical Mechanisms course, covering kinematics, dynamics and motion transmission.

Technical Skills

- **Robotics Frameworks:** ROS, ROS 2, Gazebo, RViz, Behavior Trees, State Machines.
- **Control, Localization & Navigation:** Model Predictive Control (MPC), Model Predictive Path Integral Control (MPPI), Extended Kalman Filter / Error-State Kalman Filter (EKF/ESKF), Sensor Fusion, SLAM, Path Planning, Obstacle Avoidance, Motion Control, Robot Kinematics and Dynamics, Hardware-in-the-Loop (HIL) Testing.
- **Sensors:** LiDAR, IMU, GNSS/RTK, stereo cameras.
- **Perception & Deep Learning:** PyTorch, TensorRT, YOLO (detection and segmentation), OpenCV, embedded inference on NVIDIA Jetson.
- **Programming Languages:** C++, C, Python, MATLAB, Bash, R, Java, Kotlin, SQL, VHDL.
- **Scientific Computing & Optimization:** Eigen, CasADi, NumPy, SciPy, pandas, Matplotlib, Numerical Optimization, Monte Carlo Simulation, Parameter Estimation, signal processing (time and frequency domain).
- **Development Tools & Platforms:** Linux (Ubuntu), NVIDIA Jetson, NVIDIA CUDA Toolkit, Git, GitHub Actions, Docker, CMake, Visual Studio Code.
- **Generative AI & Agents:** LLMs, agentic systems, prompt engineering.

Professional Competencies

- Technical leadership in autonomous robotics and control systems.
- Turning research algorithms into deployable, field-validated robotic systems.
- Cross-functional collaboration with international development teams (USA, India).
- Project planning, scheduling and performance benchmarking of robotics software systems.
- Technical mentoring and guidance of engineers and research collaborators.
- Effective technical communication through documentation, presentations and scientific publications.

Education

Graduate

- **São Carlos School of Engineering - University of São Paulo** **São Carlos**
Ph.D. in Mechanical Engineering 2018–2022
- **Luiz de Queiroz College of Agriculture (ESALQ) - University of São Paulo** **Piracicaba**
MBA - Data Science and Analytics 2021–2022
- **São Carlos School of Engineering - University of São Paulo** **São Carlos**
M.Sc. in Mechanical Engineering 2015–2017

Undergraduate.....

- **Federal Institute of Education, Science and Technology of São Paulo (IFSP) Araraquara**
B.Eng. in Mechanical Engineering 2017–2022
- **Federal Institute of Education, Science and Technology of São Paulo (IFSP) Araraquara**
Associate Degree in Industrial Mechatronics 2012–2014

Courses.....

- **Federal University of Rio Grande do Norte Natal**
Huawei Certification HCIA - Artificial Intelligence 2020–2021

Projects

- **LabelSketch:** Open-source, 100% client-side image annotation tool for building perception datasets, with bounding boxes and polygons and COCO/YOLO-compatible export. Part of a suite of annotation tools that also includes GeoSketch (satellite imagery) and AudioSketch (sound events). *JavaScript, HTML5 Canvas.* github.com/leo-bonacini

Publications

- **Selection of a Navigation Strategy According to Agricultural Scenarios and Sensor Data Integrity**
A framework for improving autonomous navigation by identifying, diagnosing and mitigating the limitations and failure modes of the TerraSentia robot's sensors (GNSS, LiDAR and cameras) in agricultural environments. Machine learning models (Multinomial Logistic Regression, Decision Trees, Random Forest and Gradient Boosting) select the most suitable navigation strategy based on sensor conditions, allowing the robot to keep reliable performance even when sensors degrade or fail. Click [here](#) for more information.
- **Classification of Scenery Using Multinomial Logistic Regression in a Sugarcane Crop**
A system that classifies agricultural scenes from LiDAR data collected by the Mirã 2.0 robot into four categories: *between crop rows*, *crop row on the left*, *crop row on the right* and *no crop rows*. The approach distinguishes traversable areas from regions without structured references, providing essential information for autonomous navigation and increasing the robustness of environment perception. Click [here](#) for more information.
- **GPS Signal Obstruction Identification Algorithm for Precision Agriculture**
GNSS data collected with the TerraSentia robot in a sugarcane field was used to develop a classification algorithm that detects when and where the GNSS signal is degraded or blocked, improving the reliability of GNSS-based positioning for field robots. Click [here](#) for more information.
- **Master's Dissertation: Identification of cutting parameters and material in CNC turning.**
Parallel data acquisition system for monitoring mechanical vibration, temperature and acoustic emission during turning, with signal preprocessing, time/frequency-domain feature extraction and classification of cutting parameters and material using Euclidean Distance, Artificial Neural Networks and Multinomial Logistic Regression. Click [here](#) and [here](#) for more information.